

# Review D — Macroeconomic Determinants and Trade Openness: What Moves Market Power and Misallocation?

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## Abstract

Reviews A–C established how markups, markdowns and misallocation are measured and how they interact. This review asks what *determines* them: trade policy, the exchange rate, and the macroeconomic environment. Four literatures are surveyed equation-by-equation: (i) the theory of pro-competitive trade effects, from the Lerner symmetry of protection through Melitz–Ottaviano variable markups and Atkeson–Burstein pricing-to-market; (ii) the empirical trade-liberalisation–markup literature, from the first-generation developing-country studies through the watershed of De Loecker, Goldberg, Khandelwal and Pavcnik (2016), whose input-cost channel reversed the expected sign for India; (iii) exchange-rate pass-through as a markup phenomenon, and the nascent literature connecting trade to labour-market power; and (iv) the macro determinants of misallocation — cyclical, credit conditions, and development — together with the political economy of why distortions persist, culminating in Hsieh, Klenow, Kyei Manu and Rockall (2026). A summary table places twenty studies side by side on the three dimensions that decide comparability — which policy shock is used, which object is actually measured, and where the identifying variation comes from. A final section reviews the econometric designs available for a single-country panel — exposure/shift-share identification, and the panel VAR — and states the research gap Chapter 6 fills: no study jointly estimates the response of markups, markdowns and misallocation to tariffs and exchange-rate exposure on modern Indian plant data.

## 1 Introduction and scope

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The thesis’s first two objectives measure product- and labour-market power (Review A) and misallocation (Review B) in Indian organised manufacturing, and their linkage (Review C). Those reviews answer *how much* and *how to measure*. This one asks the harder question: what *moves* these objects? When tariffs rise, when the rupee appreciates, when a state grows faster than trend, what happens to the markup  $\mu$ , the markdown  $\psi$ , and the within-industry dispersion of revenue productivity?

The question is harder for three reasons, and it is worth naming them at the outset

because they organise everything that follows. First, the theory is genuinely ambiguous. The textbook presumption — openness disciplines market power — survives only under demand systems that permit variable markups, and even then the sign can reverse once imported inputs enter the cost function. Second, the treatment is not a treatment. Tariffs are chosen, not assigned; they respond to the very industry conditions whose response we want to measure. Third, the outcome is itself an estimate. Markups and wedges arrive from Review A’s and Review B’s machinery carrying their own sampling error and their own identifying assumptions, so a regression of one estimate on another compounds two sets of caveats rather than one.

This review organises the literature in four passes: theory (§2), the trade–markup evidence from the first generation through its watershed (§§3–4), exchange rates and factor markets (§§5–6), and the macroeconomics and political economy of misallocation (§§7–8). It closes with the econometric designs available to a single-country panel and the gap Chapter 6 fills (§10). Throughout, notation follows Reviews A–B:  $\mu_{it} = \theta_{it}^M / \alpha_{it}^M$  is the production-approach markup, the ratio of the output elasticity of materials to their revenue share;  $\psi_{it} = (\theta_{it}^L / \alpha_{it}^L) / \mu_{it}$  is the labour markdown, with  $\psi > 1$  indicating wage-setting power; and  $\text{var}(\ln \text{TFPR})$  is the misallocation statistic, the within-industry variance of log revenue productivity.

## 2 Theory: why openness should discipline markups — and why it may not

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### 2.1 The CES benchmark, and why the question needs it broken

Begin with what cannot happen. Under monopolistic competition with constant elasticity of substitution (CES) demand (Krugman, 1979), every firm charges

$$\mu = \frac{\sigma}{\sigma - 1}, \quad (1)$$

where  $\sigma > 1$  is the elasticity of substitution between varieties. The markup depends on  $\sigma$  and on nothing else — not on the firm’s costs, not on its size, and crucially not on how many competitors it faces. Trade in this world reallocates output towards more productive firms and raises welfare (Melitz, 2003), but it cannot compress a single markup, because (1) has no argument for it to move. Pro-competitive effects are therefore not a robust prediction of trade theory; they are a prediction of trade theory *plus* a demand system with variable markups. Everything in this section is an account of which departure from (1) is being made, and what each one implies for the data.

### 2.2 The market-size and substitution channels

Melitz & Ottaviano (2008) deliver variable markups with linear demand. Firm  $i$ ’s optimal price over marginal cost  $c_i$  satisfies

$$p_i = \frac{1}{2} (c_i + c_D), \quad \mu_i \equiv \frac{p_i}{c_i} = \frac{1}{2} \left( 1 + \frac{c_D}{c_i} \right), \quad (2)$$

where  $c_D$  is the domestic cost cutoff — the marginal cost of the least efficient firm that survives. Opening to trade lowers  $c_D$ , because tougher selection kills the weakest incumbents, and every survivor’s markup is compressed: the **pro-competitive effect**.

Three features of (2) matter empirically. First, the effect is a *level* effect on the whole distribution, and it is strongest for the firms with the lowest  $c_i$  — that is, for the high-markup firms, since  $\mu_i$  is decreasing in  $c_i$ . Trade compresses the right tail of the markup distribution more than the middle, so a study that reports only mean markups will understate it. Second, the channel runs through import competition in the *output* market, so it should load on output tariffs and on import-exposure interactions, not on input tariffs. Third, the mechanism is selection, which means it operates through entry and exit as much as through incumbent repricing; in a short panel with limited turnover, the measurable part is the incumbent margin alone.

The presumption is not safe even here. Arkolakis et al. (2019) show that once the demand system is disciplined by the same data used to calibrate the gains from trade, the pro-competitive effect can shrink to nothing or reverse: markups fall for some firms and rise for others, and the aggregate welfare contribution of the markup channel is close to zero in the leading parameterisations. Their point is not that markups do not move. It is that the *direction of the aggregate* is a quantitative question in which the dispersion of markups matters more than the level — which is precisely why this thesis measures dispersion alongside means, and why Review C’s markup–misallocation linkage is not a side issue.

### 2.3 Pricing-to-market and the markup elasticity

Atkeson & Burstein (2008) generate variable markups from a different departure: nested CES with oligopolistic competition, so that a firm large enough to affect its sector’s price index internalises that effect. Firm  $i$ ’s markup depends on its market share  $s_i$ ,

$$\mu_i = \frac{\varepsilon(s_i)}{\varepsilon(s_i) - 1}, \quad \varepsilon(s_i) = \left[ s_i \frac{1}{\eta} + (1 - s_i) \frac{1}{\rho} \right]^{-1}, \quad (3)$$

where  $\rho$  is the elasticity of substitution *within* a sector,  $\eta$  the elasticity *across* sectors, and  $\rho > \eta$  so that varieties inside a sector are closer substitutes than sectors are for one another. As  $s_i \rightarrow 0$  the perceived elasticity  $\varepsilon(s_i) \rightarrow \rho$  and (3) collapses to the CES benchmark (1); as  $s_i \rightarrow 1$  it tends to  $\eta$ , and the markup is at its highest.

Because  $\partial \mu_i / \partial s_i > 0$ , any shock that reallocates market shares — a tariff, an exchange-rate movement, the entry of an importer — moves markups mechanically, without any firm changing its pricing rule. The *markup elasticity*

$$\Gamma_i \equiv \frac{\partial \ln \mu_i}{\partial \ln s_i} \quad (4)$$

is the sufficient statistic for how much: it converts a reallocation of shares into a movement in markups. It is the oligopoly counterpart of the demand-side superelasticity that Chapter 5 estimates for India (0.183), and the two are alternative parameterisations of the same curvature. Amiti et al. (2019) show that  $\Gamma_i$  also governs strategic complementarity in price setting: firms with larger shares respond more to their competitors’ prices, so an import shock propagates to purely domestic firms through the sector price index even when those firms import nothing. That indirect channel is invisible to a design that compares importers with non-importers, and it is a reason to prefer industry-level exposure measures.

## 2.4 Lerner symmetry and the input-cost channel

Lerner (1936) taught that an import tariff is an export tax: protection raises the domestic relative price of importables *and* the cost of imported inputs. For measured markups the second margin is decisive, and it is the single most important idea in this review.

Write marginal cost as  $c_{it} = c(w_t, p_{it}^M(\tau_{st}^{inp}))$ , a function of the wage  $w_t$  and of the price of intermediates  $p_{it}^M$ , which depends on the input tariff  $\tau_{st}^{inp}$  facing industry  $s$ . Let  $\beta$  be the pass-through of cost to price,  $\Delta \ln p = \beta \Delta \ln c$ . Then

$$\Delta \ln \mu_{it} = \Delta \ln p_{it} - \Delta \ln c_{it} = -(1 - \beta) \Delta \ln c_{it} > 0 \quad \text{when } \Delta \ln c_{it} < 0, \quad (5)$$

so that whenever pass-through is incomplete ( $\beta < 1$ ), a fall in costs *raises* the measured markup. Liberalisation that cheapens imported inputs therefore pushes markups up, in direct opposition to the pro-competitive compression of (2). The two channels are not alternatives to be adjudicated in theory; they are simultaneous, they carry opposite signs, and which one dominates is the empirical content of §4.

Equation (5) is also the reading key for the episode this thesis studies. India’s post-2012 decade is one of *rising* protection, not falling, so the channel runs in reverse: tariff increases raise input costs, and with  $\beta < 1$  they compress measured markups. A finding that protection lowers markups is thus not evidence against market power. It is the cost-channel prediction, and distinguishing it from a genuine competitive effect requires the output- and input-tariff margins to be separated. Halpern et al. (2015) quantify how large the input margin can be: imported inputs account for roughly a quarter of Hungarian firms’ productivity growth, with the gain arising from variety and quality rather than price alone.

## 2.5 From markups to misallocation in theory

The channels above concern the markup *level*. The thesis’s third object is dispersion, and the theoretical link runs through Review C: when markups differ across firms facing the same factor prices, marginal revenue products differ too, and the wedge accounting of Review B records the difference as misallocation. Epifani & Gancia (2011) make the trade connection explicit — openness changes the *distribution* of markups, not just its mean, and a trade liberalisation that raises markup dispersion can lower welfare even while raising average productivity. Edmond et al. (2015) quantify the same mechanism in a calibrated economy and find the misallocation channel economically large relative to the conventional gains. Feenstra & Weinstein (2017) reach the level from the other direction, showing that once variable markups are admitted, import competition delivers a welfare gain through markup compression that a CES accounting misses entirely. Together these papers justify treating markup dispersion as an outcome in its own right, which is what Chapter 6 does.

## 3 First-generation evidence: liberalisation disciplines margins

The early plant-data literature asked the question directly, by regressing price–cost margins on openness. Levinsohn (1993), using Turkey’s 1984 liberalisation, and Harrison (1994), using Côte d’Ivoire’s 1985 reform, both found margins falling in the sectors that were opened — the “imports-as-market-discipline” hypothesis, and the first systematic

evidence for (2). For India, [Krishna & Mitra \(1998\)](#) exploited the 1991 reform with the canonical difference-in-differences specification

$$\text{margin}_{jt} = \alpha_j + \gamma_t + \delta \text{Reform}_t \times \text{Ind}_j + \varepsilon_{jt}, \quad (6)$$

where  $\text{margin}_{jt}$  is industry  $j$ 's accounting price–cost margin in year  $t$ ,  $\alpha_j$  and  $\gamma_t$  are industry and year effects,  $\text{Reform}_t$  switches on after 1991,  $\text{Ind}_j$  measures industry  $j$ 's exposure to it, and  $\delta$  is the effect of interest. They found markups falling and productivity growth rising after 1991.

These studies share two limitations, and Review A's toolkit was built to fix the first of them. Margins in (6) are accounting ratios — revenue minus measured variable cost, over revenue — not estimated markups. They therefore inherit every distortion in the measurement of variable cost, they cannot separate a markup from a returns-to-scale effect, and they say nothing about the labour margin. The second limitation is identification: tariff endogeneity was addressed only through the suddenness of the reform, which is an appeal to surprise rather than to exogeneity.

[Topalova \(2010\)](#) and [Topalova & Khandelwal \(2011\)](#) sharpened the Indian design considerably. The 1991–97 cuts were imposed externally as a condition of IMF support, applied according to a formula that compressed tariffs towards a uniform rate, and therefore bore little relation to the lobbying position of individual industries — an unusually credible source of variation. Measuring tariffs at the industry  $\times$  year level and using TFP as the outcome, they found that lower output tariffs raised competitive pressure while lower *input* tariffs raised firm productivity through cheaper and better inputs. [Goldberg et al. \(2010\)](#) pushed the input channel further, showing that access to new imported varieties expanded Indian firms' product scope, and [Ahsan \(2013\)](#) showed the input-tariff effect is mediated by contract enforcement, so the same tariff cut delivers different gains in different states. [Nataraj \(2011\)](#) added the distributional caveat that matters for a formal-sector study like this one: input-tariff cuts raised productivity in formal firms but the informal sector responded differently, so estimates from the organised sector do not generalise to manufacturing as a whole. [De Loecker & Goldberg \(2014\)](#) and [Goldberg & Pavcnik \(2016\)](#) survey the resulting consensus, which is that the input margin is at least as important as the output margin and frequently larger.

## 4 The watershed: De Loecker, Goldberg, Khandelwal and Pavcnik (2016)

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[De Loecker et al. \(2016\)](#) is the paper that reorganised the field, and it did so by removing the measurement compromise that (6) rested on. They combine the production-approach markup of Review A with plant-level *prices* from Indian Prowess data, which lets them decompose an observed price movement into its two components,

$$\ln p_{ijt} = \ln \mu_{ijt} + \ln c_{ijt}, \quad \begin{cases} \Delta \tau_{st}^{\text{out}} \downarrow \Rightarrow p \downarrow & (\text{competition}) \\ \Delta \tau_{st}^{\text{inp}} \downarrow \Rightarrow c \downarrow\downarrow & (\text{input costs}), \end{cases} \quad (7)$$

where  $p_{ijt}$  is the price of product  $j$  made by firm  $i$ ,  $\mu_{ijt}$  its markup,  $c_{ijt}$  its marginal cost, and  $\tau^{\text{out}}$ ,  $\tau^{\text{inp}}$  the output and input tariffs facing the firm's industry  $s$ . Having prices *and* estimated markups means the two right-hand terms are separately observed rather than one being inferred as a residual.

Their central estimates reversed the expected sign. Input-tariff cuts lowered marginal costs by far more than prices fell — factory-gate prices declined about 18 per cent, but costs declined about 31 per cent — so **markups rose** with liberalisation. The pro-competitive compression from output tariffs was present and correctly signed, but it was swamped by incomplete cost pass-through, exactly as (5) predicts. Indian trade liberalisation made firms more efficient and left them collecting a larger margin.

Three lessons carry into this thesis. (i) Output and input tariffs are different treatments with opposite predicted signs and must be measured separately; a single “tariff” regressor identifies their difference, not either one. Chapter 6 measures output tariffs and flags the input-tariff panel, which requires input–output weights, as the natural extension. (ii) The sign of the markup response to protection is an empirical question rather than a theoretical presumption — a discipline this review tries to maintain throughout. (iii) Identification came from industry  $\times$  year tariff variation with firm and year effects, which is the exposure design §10 formalises.

There is also a limitation that bears directly on the data used here, and it should be stated plainly rather than buried. DLGKP’s decomposition requires plant-level prices, and the Annual Survey of Industries does not report them. Markups estimated from ASI are revenue-based, so the price and markup movements in (7) cannot be separated on this data: what Chapter 6 estimates is the response of the *revenue-based* markup, which bundles the two. This is a real constraint on interpretation, not a technicality, and it is why the chapter leans on the sign pattern across output-tariff and exchange-rate margins rather than on any single coefficient.

## 5 Exchange rates as competition policy: pass-through and markups

### 5.1 Pass-through is a markup phenomenon

The exchange rate is a tariff that nobody legislates. A real appreciation lowers the domestic-currency price of every importable, which is precisely what an output-tariff cut does, and it does so without passing through parliament or responding to a lobby. That makes it an attractive second margin for a study whose first margin — the tariff — is endogenous by construction.

The canonical exchange-rate pass-through (ERPT) regression is

$$\Delta \ln p_{it}^x = \beta \Delta \ln e_t + \mathbf{x}_{it}' \gamma + \varepsilon_{it}, \quad (8)$$

where  $p_{it}^x$  is firm  $i$ ’s export price in destination currency,  $e_t$  the nominal exchange rate,  $\mathbf{x}_{it}$  a vector of controls, and  $\beta$  the pass-through coefficient. The literature finds  $\beta < 1$  with great regularity (Goldberg & Knetter, 1997). The interpretation is the point: under constant markups, cost shocks would pass one-for-one into price and  $\beta$  would equal one, so *incomplete pass-through is markup adjustment*. Equation (8) and equation (5) are the same statement written for different shocks.

The firm-level evidence sharpens this. Berman et al. (2012) show that high-performance French exporters respond to a depreciation by raising their markups rather than cutting their destination prices — pass-through falls with firm productivity, exactly as (3) implies when large firms have larger markup elasticities. Amiti et al. (2014) close the loop by adding the cost side: import-intensive exporters face a depreciation that raises their

input costs at the same time as it improves their competitive position, and the two effects reinforce to produce doubly low pass-through. [Burstein & Gopinath \(2014\)](#) survey the resulting picture, in which the currency of invoicing, the import intensity of the exporter, and the competitive structure of the destination market all enter the same equation.

For a study of the *domestic* market the argument runs symmetrically, and this is the version Chapter 6 uses. An appreciation lowers the home-currency price of rivals' imports, squeezing domestic producers in import-exposed industries; by (5) those producers absorb part of the squeeze in their markups rather than passing it fully into price. The prediction is that appreciation compresses markups where import exposure is high and leaves them alone where it is not — which is the sign Chapter 6 finds for India at both NIC-2 and NIC-3 levels of industry detail. Because domestic firms also buy imported intermediates, the appreciation simultaneously cheapens their inputs, so the net effect is again a race between the competition and cost channels; the exposure interaction is what separates them.

## 5.2 From national series to identification

The real effective exchange rate is a single national series, which means it is perfectly collinear with a full set of year effects: no within-country study can identify its effect from the time-series alone without assuming away every other national shock. The literature's answer is to identify from *exposure interactions*, differencing a common shock across predetermined cross-sectional exposures. The generic estimating equation,

$$Y_{st} = \beta \left( \text{REER}_t - \overline{\text{REER}} \right) \times s_{s,0}^{\text{imp}} + \alpha_s + \delta_t + \varepsilon_{st}, \quad (9)$$

regresses the outcome  $Y_{st}$  for industry  $s$  in year  $t$  on the demeaned national REER interacted with  $s_{s,0}^{\text{imp}}$ , industry  $s$ 's import share frozen at a base year, with industry effects  $\alpha_s$  and year effects  $\delta_t$ . The year effects absorb the REER's own main effect along with every other aggregate;  $\beta$  is identified purely from the differential response of more- and less-exposed industries. This is a shift-share, or Bartik, design ([Bartik, 1991](#)), and what it estimates is a *relative* response — the honest object for a single country, and one whose limits §10 sets out.

## 6 Trade, macro conditions, and labour-market power

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The markdown literature of Review A and the trade literature of this one are only beginning to meet, and the thinness of the intersection is itself the finding. Most of what exists approaches from the input side. [Morlacco \(2020\)](#) estimates the buyer power of French importers over their foreign suppliers using the two-flexible-input wedge — formally the same object as  $\psi$ , applied to imported intermediates rather than to labour — and shows that openness and factor-market power are determined jointly. [Rubens \(2023\)](#) shows that oligopsony power distorts input allocation in Chinese tobacco, connecting markdowns directly to the misallocation statistic and demonstrating that the two thesis objects are not independent. [Dobbelaere & Mairesse \(2013\)](#) provide the methodological bridge, estimating product- and labour-market imperfections jointly from the production function and documenting how often the two coexist within the same industry.

For developing-country labour markets specifically, [Amodio & de Roux \(2023\)](#) document substantial plant-level monopsony in Colombia and find wedges rising as import competition intensifies, and [Brooks et al. \(2021\)](#) supply the India–China markdown benchmarks

this thesis extends. [Manning \(2003\)](#) provides the theoretical frame within which these results are read: when search frictions or thin local markets give employers an upward-sloping labour supply curve, the wage is set below the marginal revenue product, and anything that shifts labour demand shifts the wedge.

The theoretical presumption here is genuinely two-sided, which is why the empirical question is open. Protection props up product demand and therefore labour demand, which in a thin local market erodes the employer’s wage-setting power by forcing it to compete harder for workers; import competition does the opposite, contracting demand and restoring the employer’s position. Against this, a protected firm earning higher rents may simply share fewer of them, and a firm squeezed by imports may cut employment rather than wages. Chapter 6’s finding that tariffs at fine industry detail *reduce* monopsony incidence, while appreciation weakly raises it, is consistent with the first mechanism and is — to the best of this review’s knowledge — the first plant-panel estimate of the trade–monopsony margin for India.

## 7 Macroeconomic determinants of misallocation

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### 7.1 Cycles, credit and crises

Misallocation is not acyclical, and financial frictions are where most of its movement is concentrated. [Gilchrist et al. \(2017\)](#) model dispersion in borrowing costs as an interest-rate wedge and show that a credit crunch raises the dispersion of the marginal revenue product of capital wherever collateral constraints bind — misallocation rises in a recession not because policy worsened but because the constraint tightened. [Midrigan & Xu \(2014\)](#) and [Moll \(2014\)](#) formalise the general-equilibrium version and reach a qualified conclusion: financial frictions generate large static misallocation, but self-financing lets productive firms outgrow the constraint over time, so the long-run losses are considerably smaller than the one-period accounting implies. [Buera et al. \(2011\)](#) trace the same friction across sectors, and [Gopinath et al. \(2017\)](#) supply the cleanest identified case, showing capital misallocation in southern Europe *rising* through a decade of cheap capital inflows because the inflows were allocated by collateral rather than by productivity.

For India specifically, [Bau & Matray \(2023\)](#) exploit the staggered liberalisation of foreign capital access across industries and show that easing it *reduced* capital misallocation, with MRPK dispersion falling in treated industries. That result establishes the point Chapter 6 builds on: Indian misallocation does respond to policy, and it responds to factor-market policy in particular. [Asker et al. \(2014\)](#) supply the permanent caution against over-reading any of this — volatility combined with capital adjustment costs generates TFPR dispersion mechanically, in an economy with no distortions at all, so cyclical dispersion is not by itself evidence of a worsening distortion.

### 7.2 Development and the wedge distribution

Across countries, dispersion falls with income ([Hsieh & Klenow, 2009](#); [Restuccia & Rogerson, 2017](#); [Bartelsman et al., 2013](#)): rich economies put their resources in better hands, and the gap is large enough to account for a substantial share of cross-country income differences. Within a single developing country over time, however, the sign is unsettled and the cross-country regularity offers little guidance. Growth episodes attract heterogeneous entrants and expand constrained incumbents unevenly, so measured dispersion

can *rise* along a transition path even while the policy environment improves. This is the reading the thesis gives to its state-level finding that Indian states growing faster than trend see TFPR dispersion fall by less than slower-growing states — not a paradox, but the expected signature of reallocation in progress. The political-economy result of §8 supplies the complementary reason why the measured wedges do not simply erode.

## 8 Why distortions persist: political economy

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Grossman & Helpman (1994) formalise protection-for-sale: tariffs are the equilibrium outcome of a bargain between organised industry interests and a government that values both contributions and welfare. The empirical content is that trade policy is an *outcome* of industry conditions — markups, employment, distress — as much as a cause of them, and Goldberg & Maggi (1999) confirm the model’s structure holds quantitatively on US data. Bown & Tovar (2011) document the Indian case directly: after the 1990s tariff reform, antidumping duties and safeguards partially restored protection precisely where liberalisation had bitten hardest, so the measured tariff schedule understates how much protection was actually unwound.

The econometric implication is first-order rather than a caveat. A single-equation regression of market power on tariffs faces reverse causality of exactly the kind the theory predicts, which motivates the two responses §10 develops: predetermined exposure designs that do not use contemporaneous policy variation, and explicit modelling of the feedback. Chapter 6’s panel VAR provides direct evidence that the concern is real — markups and misallocation Granger-cause tariffs in 2012–2022 India, which is the protection-for-sale mechanism showing up in the data at annual frequency.

Hsieh et al. (2026) close the circle at the level of welfare. Removing India’s size-dependent wedges would raise output by 1–4 per cent but *lower* consumption-equivalent welfare by roughly 9 per cent, because the losses fall on low-skill workers and small-firm owners while the gains accrue to the owners of the firms that expand. A majority can therefore rationally oppose a reform that raises measured output. Distortions persist because their removal is distributionally unattractive, and protection persists for the same reason — which reframes the thesis’s misallocation estimates as a measure of what reform would cost as much as what it would gain.

## 9 The empirical record at a glance

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Table 1 places the studies surveyed above side by side. Three columns do most of the work, and they are the ones this literature most often leaves implicit.

The first is the *policy shock*. “Trade liberalisation” is not one treatment but at least four — an output-tariff cut, an input-tariff cut, a non-tariff barrier, a capital-account opening — and (5) shows that the first two carry opposite predicted signs for the measured markup. A study that regresses a margin on an aggregate openness index is estimating a weighted average of effects that partly cancel, which is a large part of why the first-generation results in §3 and the De Loecker et al. (2016) results in §4 appear to disagree. They mostly do not disagree; they are answering different questions.

The second is the *outcome measured*. An accounting price–cost margin, an estimated markup, a marginal cost and a factor wedge are four different objects, and only the last

two require the production-function machinery of Review A. The column makes visible how recent the ability to measure markdowns is: not one study before 2020 reports one, which is the mechanical reason the trade–monopsony margin is nearly empty.

The third is *identification*. The Indian studies cluster on the 1991–97 tariff cuts precisely because their externally-imposed, formula-driven character made them credibly exogenous — a source of variation that does not exist for the post-2012 period this thesis studies, where tariffs were set domestically and, by §8, endogenously. That absence is why Chapter 6 leans on predetermined exposure interactions and on an explicit feedback model rather than claiming a natural experiment it does not have.

## 10 Econometric designs for a single-country panel, and the research gap

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### 10.1 Exposure (shift-share) designs

Equation (9) generalises to any national shock  $g_t$  interacted with a predetermined exposure  $s_{s,0}$ , the structure Autor et al. (2013) made standard. Identification can be defended in either of two ways, and they make different demands. Goldsmith-Pinkham et al. (2020) show the design is equivalent to using the *shares* as instruments, so validity rests on the exposure shares being exogenous to the outcome’s trend — a strong assumption when shares reflect historical comparative advantage. Borusyak et al. (2022) take the opposite route, deriving conditions under which many quasi-random *shocks* deliver consistency even when individual shares are endogenous. Adão et al. (2019) show that conventional standard errors over-reject badly in these designs because observations sharing similar exposure are effectively correlated, and that inference should be conducted at the level of the exposure rather than the unit.

In Chapter 6 the shares are 2012 import shares, predetermined by construction relative to the 2012–2022 outcomes; the shocks are the REER path and the industry tariff schedule; and year effects absorb all national aggregates. The design therefore estimates relative responses across industries, and the absolute effect of a national shock on the national level of markups is not identified — a limitation worth stating once, clearly, rather than letting a reader infer more than the design supports.

### 10.2 Panel vector autoregression

When the question involves dynamics and feedback rather than the response to a single shock, and when units are heterogeneous, the panel VAR of Love & Zicchino (2006) and Abrigo & Love (2016) is the natural instrument:

$$\mathbf{y}_{it} = \mathbf{A}(L) \mathbf{y}_{it-1} + \mathbf{u}_i + \mathbf{e}_{it}, \quad (10)$$

where  $\mathbf{y}_{it}$  stacks the endogenous variables for unit  $i$  in year  $t$ ,  $\mathbf{A}(L)$  is a matrix lag polynomial,  $\mathbf{u}_i$  are unit fixed effects and  $\mathbf{e}_{it}$  the innovations. Estimating (10) by GMM after eliminating  $\mathbf{u}_i$  delivers Granger-causality tests and orthogonalised impulse responses. The elimination cannot be done by within-transformation: in a dynamic panel that induces the correlation between the transformed regressor and error described by Nickell (1981), whose bias is of order  $1/T$  and therefore serious at the sample lengths available here. The standard remedy is the forward orthogonal deviations, or Helmert, transform

Note how late mark downs enter the table — the trade-monopsony margin is empty not because it was settled but because the object could not be measured. Cells marked n.r. are not reported in a form this review can state confidently.  $PCM$  = accounting price-cost margin;  $PF$  = production-function;  $DiD$  = difference-in-differences;  $MRPK$  = marginal revenue product of capital.

| Study                                                                   | Setting and data                                               | Period           | Policy shock                                                                          | Outcome measured                                            | Identification                                                       | Key finding                                                                                                                                                        |
|-------------------------------------------------------------------------|----------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Panel A. First-generation: margins and openness</i>                  |                                                                |                  |                                                                                       |                                                             |                                                                      |                                                                                                                                                                    |
| Levinsohn (1993)                                                        | Turkey, plant data                                             | 1983–86          | 1984 liberalisation                                                                   | PCM                                                         | DiD across sectors                                                   | Margins fell where trade was opened: the imports-as-market-discipline result                                                                                       |
| Harrison (1994)                                                         | Côte d'Ivoire, plants                                          | 1979–87          | 1985 reform                                                                           | PCM and TFP                                                 | Panel, sector exposure                                               | Margins fell and productivity rose after the reform                                                                                                                |
| Krishna & Mitra (1998)                                                  | India, industry panel                                          | around 1991      | 1991 reform                                                                           | PCM, productivity growth                                    | DiD, (6)                                                             | Markups fell and productivity growth rose after 1991                                                                                                               |
| <i>Panel B. The Indian tariff literature: the input channel emerges</i> |                                                                |                  |                                                                                       |                                                             |                                                                      |                                                                                                                                                                    |
| Topalova (2010)                                                         | India, districts and firms                                     | 1987–2001        | 1991–97 tariff cuts                                                                   | Poverty, TFP                                                | Externally imposed, formula-driven cuts                              | Factor immobility concentrates the adjustment regionally                                                                                                           |
| Topalova & Khandelwal (2011)                                            | India, firms (Prowess)                                         | 1989–2003        | Output <i>and</i> input tariffs                                                       | TFP                                                         | Industry $\times$ year tariffs, firm effects                         | Input-tariff cuts raise productivity by more than output-tariff cuts                                                                                               |
| Goldberg et al. (2010)                                                  | India, firms                                                   | 1989–2003        | Input tariffs                                                                         | Product scope, input variety                                | Exposure design                                                      | Access to imported varieties expands firms' product scope                                                                                                          |
| Ahsan (2013)                                                            | India, firms                                                   | n.r.             | Input tariffs                                                                         | TFP                                                         | Tariff $\times$ contract enforcement                                 | The input-tariff gain is mediated by state institutional quality                                                                                                   |
| Nataraj (2011)                                                          | India, formal <i>and</i> informal                              | n.r.             | Output and input tariffs                                                              | TFP                                                         | Exposure design                                                      | Formal and informal sectors respond differently: organised-sector estimates do not generalise                                                                      |
| <i>Panel C. The watershed, and the exchange-rate margin</i>             |                                                                |                  |                                                                                       |                                                             |                                                                      |                                                                                                                                                                    |
| De Loecker et al. (2016)                                                | India, Prowess with plant <i>prices</i>                        | 1989–2003        | Output and input tariffs                                                              | Markup <i>and</i> marginal cost, separately                 | PF markups plus observed prices                                      | Costs fell $\approx 31\%$ , prices $\approx 18\%$ : markups <b>rose</b> with liberalisation                                                                        |
| Goldberg & Knetter (1997)                                               | Multi-country, survey                                          | —                | Exchange rate                                                                         | Pass-through $\beta$                                        | Survey of ERPT designs                                               | $\beta < 1$ is pervasive; incomplete pass-through <i>is</i> markup adjustment                                                                                      |
| Berman et al. (2012)                                                    | France, exporters                                              | n.r.             | Depreciation                                                                          | Export price, markup                                        | Firm-level ERPT                                                      | High-productivity exporters absorb the shock in markups, not prices                                                                                                |
| Amiti et al. (2014)                                                     | Belgium, firms                                                 | n.r.             | Exchange rate                                                                         | Export price                                                | ERPT with import intensity                                           | Import-intensive exporters have doubly low pass-through                                                                                                            |
| Amiti et al. (2019)                                                     | Belgium, firms                                                 | n.r.             | International cost shocks                                                             | Domestic prices                                             | Strategic complementarity                                            | The markup elasticity (4) governs how shocks propagate to non-importers                                                                                            |
| <i>Panel D. Factor-market power, and misallocation as an outcome</i>    |                                                                |                  |                                                                                       |                                                             |                                                                      |                                                                                                                                                                    |
| Morlacco (2020)                                                         | France, importers                                              | 1996–2007        | Openness (input side)                                                                 | Buyer-power wedge                                           | Two-flexible-input wedge                                             | Buyer power over foreign suppliers is large and varies with openness                                                                                               |
| Amodio & de Roux (2023)                                                 | Colombia, plants                                               | 1982–2012        | Import competition                                                                    | <b>Markdown</b>                                             | PF, two-input ratio                                                  | Substantial plant monopsony; wedges rise as import competition intensifies                                                                                         |
| Rubens (2023)                                                           | China, tobacco                                                 | n.r.             | Market structure                                                                      | Oligopsony power, misallocation                             | Structural model                                                     | Oligopsony power distorts input allocation directly                                                                                                                |
| Bau & Matray (2023)                                                     | India, firms                                                   | n.r.             | FDI / capital-account liberalisation                                                  | MRPK dispersion                                             | Staggered industry roll-out                                          | Capital-market integration <b>reduced</b> misallocation                                                                                                            |
| Gopinath et al. (2017)                                                  | Spain and southern Europe                                      | 1999–2012        | Capital inflows                                                                       | MRPK dispersion                                             | Wedge accounting plus a frictions model                              | Misallocation <b>rose</b> as cheap capital was allocated by collateral                                                                                             |
| Bown & Tovar (2011)                                                     | India                                                          | post-1991        | Antidumping, safeguards                                                               | Realised protection                                         | Policy-instrument audit                                              | Protection was partly restored where liberalisation bit hardest                                                                                                    |
| Hsieh et al. (2026)                                                     | India, formal and informal                                     | 2000–2012        | Size-dependent wedges                                                                 | Output, welfare                                             | Wedge accounting with skill heterogeneity                            | Output +1–4%, but consumption-equivalent welfare –9%                                                                                                               |
| <b>This thesis (Ch. 6)</b>                                              | <b>India, ASI plant panel aggregated to industry and state</b> | <b>2012–2022</b> | <b>Output tariffs (NIC-2008 concordance) and REER <math>\times</math> 2012 import</b> | <b>Markup, markdown <i>and</i> TFPR dispersion, jointly</b> | <b>Predetermined exposure interactions plus a stability-verified</b> | <b>Protection compresses markups (the cost channel in reverse); appreciation disciplines exposed industries; monopsony falls with protection; misallocation is</b> |

of [Arellano & Bover \(1995\)](#), which subtracts the mean of all *future* observations and so preserves orthogonality to the lagged regressors.

Two practice notes this thesis validates. Common time effects must be removed first, by year-demeaning, or trending panels violate the stability condition that the eigenvalues of the companion matrix lie inside the unit circle. And lag order should be selected by the moment and model selection criteria of [Andrews & Lu \(2001\)](#) rather than by the information criteria familiar from time-series VARs, which are not valid under GMM estimation. With  $T = 11$  the horizon is short, and the resulting impulse responses should be read as descriptions of predictive dynamics rather than as structural shocks — a distinction Chapter 6 maintains in its wording.

### 10.3 The research gap

Across §§3–8 no study estimates, on modern Indian plant data, the joint response of the three thesis objects — markups, markdowns/monopsony, and misallocation — to trade policy and exchange-rate exposure, with the political-economy feedback made explicit. The 1990s liberalisation literature stops at productivity or, in the case of [De Loecker et al. \(2016\)](#), at markups, before markdowns were measurable at all. The markdown literature has not met Indian trade policy. The Indian branch of the misallocation literature ([Bau & Matray, 2023](#); [Hsieh et al., 2026](#)) studies capital-market and size-dependent policy rather than the border. And the three objects have never been placed in the same panel, which matters because Review C establishes that they are not independent: a markup response mechanically moves the misallocation statistic, so estimating them separately risks reporting one effect twice.

Chapter 6 fills this gap with an eleven-year industry and state panel carrying all three objects, a newly concorded NIC-2008 tariff panel (2011–2023), REER-exposure interactions, state accounts, and a stability-verified panel VAR. It finds protection compressing markups — the [De Loecker et al. \(2016\)](#) cost channel of (5) running in reverse — appreciation disciplining exposed industries, monopsony falling with protection, misallocation orthogonal to the border but rising with state growth, and tariffs responding to industry conditions in the manner [Grossman & Helpman \(1994\)](#) predicts.

## 11 Debates, open questions, and the bridge to Chapter 6

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Four questions remain genuinely open, and each shapes a design choice in what follows.

**Does openness compress markups at all?** The honest answer is that it depends on which margin is opened and on what is being measured. The pro-competitive channel of (2) is real and correctly signed wherever output markets are opened, but [Arkolakis et al. \(2019\)](#) show its aggregate welfare contribution can vanish once the demand system is disciplined, and [De Loecker et al. \(2016\)](#) show the input channel can reverse the sign of the measured response entirely. The literature’s apparent disagreement is largely an artefact of pooling treatments. This review’s position is that the output and input margins should never be estimated as one regressor, and that where only one can be measured — as in Chapter 6, which has output tariffs — the omission should be stated rather than absorbed into the interpretation.

**Is the revenue-based markup good enough?** Every estimate in Panels A–C of

Table 1 except [De Loecker et al. \(2016\)](#) is revenue-based, and (7) shows what that costs: price and markup movements cannot be separated without plant prices, which the Annual Survey of Industries does not collect. The response taken here follows Reviews A and B — privilege comparative statements over levels. A revenue-based markup that is contaminated in the same way in every year and every industry still supports a credible statement about how it *responds* to a shock that varies across industries and years, which is precisely what an exposure design estimates.

**Does trade move labour-market power?** This is the emptiest cell in the table, and the theory is two-sided (§6). Only [Amodio & de Roux \(2023\)](#) address it directly, on Colombian data, and the markdown literature has not yet met Indian trade policy at all. The gap is an artefact of sequencing: markdowns became routinely measurable only after [Brooks et al. \(2021\)](#), by which time the great liberalisation episodes were two decades past. A country experiencing a *rising*-protection decade with measurable markdowns is therefore an unusually direct test, and that is the setting Chapter 6 occupies.

**Is misallocation a macro outcome or a structural constant?** The evidence in §7 says clearly that it moves — [Bau & Matray \(2023\)](#) and [Gopinath et al. \(2017\)](#) both find large policy-driven movements, in opposite directions — but [Asker et al. \(2014\)](#) establish that some of the movement is adjustment rather than distortion, and neither of those studies uses trade policy as the shock. Whether the border moves misallocation in India is, as far as this review can establish, untested.

Taken together these four questions define the object Chapter 6 must be: a single panel carrying markups, markdowns and misallocation on consistent definitions; two distinct sources of variation, one legislated and endogenous (tariffs), one national and exogenous to any single industry but usable only through exposure (the REER); and an explicit model of the feedback from industry conditions back to policy, because §8 predicts it and a single-equation design would be biased by it. Review C explains why the three objects cannot be estimated independently of one another; this review explains why they cannot be regressed naively on policy. Chapter 6 is the response to both constraints.

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